

Array Decomposition

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TBD

Table of contents

1	Introduction	3
2	Decomposition	3
3	Projectors	3
4	Least Squares	3
5	Interactions	3
6	Log-linear Analysis	3
7	Orthogonal functions	3

Note: This is a working manuscript which will be expanded/updated frequently. All suggestions for improvement are welcome. All Rmd, tex, html, pdf, R, and C files are in the public domain. Attribution will be appreciated, but is not required. The files can be found at <https://github.com/deleeuw>

1 Introduction

2 Decomposition

$$a_i = a_{\bullet} + (a_i - a_{\bullet})$$

$$a_{ij} = a_{\bullet\bullet} + (a_{i\bullet} - a_{\bullet\bullet}) + (a_{\bullet j} - a_{\bullet\bullet}) + (a_{ij} - a_{i\bullet} - a_{\bullet j} + a_{\bullet\bullet})$$

$$\begin{aligned} a_{ijk} = a_{\bullet\bullet\bullet} &+ (a_{i\bullet\bullet} - a_{\bullet\bullet\bullet}) + (a_{\bullet j\bullet} - a_{\bullet\bullet\bullet}) + (a_{\bullet\bullet k} - a_{\bullet\bullet\bullet}) + \\ &(a_{ij\bullet} - a_{i\bullet\bullet} - a_{\bullet j\bullet} + a_{\bullet\bullet\bullet}) + \\ &(a_{i\bullet k} - a_{i\bullet\bullet} - a_{\bullet\bullet k} + a_{\bullet\bullet\bullet}) + \\ &(a_{\bullet jk} - a_{\bullet j\bullet} - a_{\bullet\bullet k} + a_{\bullet\bullet\bullet}) + \\ &(a_{ijk} - a_{ij\bullet} - a_{i\bullet k} - a_{\bullet jk} + a_{i\bullet\bullet} + a_{\bullet j\bullet} + a_{\bullet\bullet k} - a_{\bullet\bullet\bullet}) \quad (1) \end{aligned}$$

3 Projectors

$$\{P_1(A)\}_{ijk} = a_{i\bullet\bullet}$$

$$\{Q_1(A)\}_{ijk} = \delta^{ijk} - a_{i\bullet\bullet}$$

$$\{P_{12}(A)\}_{ijk} = a_{ij\bullet}$$

4 Least Squares

5 Interactions

6 Log-linear Analysis

7 Orthogonal functions

Suppose M_n is the set of matrices X of order n with

- first column all elements equal to one, and
- $X'X = nI$.

Suppose the array A is $n \times m \times \ell$. Choose $X \in M_n$, $Y \in M_m$, and $Z \in M_\ell$. Define

$$\alpha_{ijk} = \sum_{i'=1}^n \sum_{j'=1}^m \sum_{k'=1}^l a_{i'j'k'} x_{i'i} y_{j'j} z_{k'k}$$